Moisture-Based Automated Watering System

The "Moisture-Based Automated Watering System" project explores a practical and efficient solution for automated irrigation, leveraging sensor-based technology to optimize water usage and ensure consistent plant care. This system is designed to monitor soil moisture levels in real-time and automatically activate a watering mechanism when moisture falls below a predefined threshold, ensuring precise and timely irrigation.

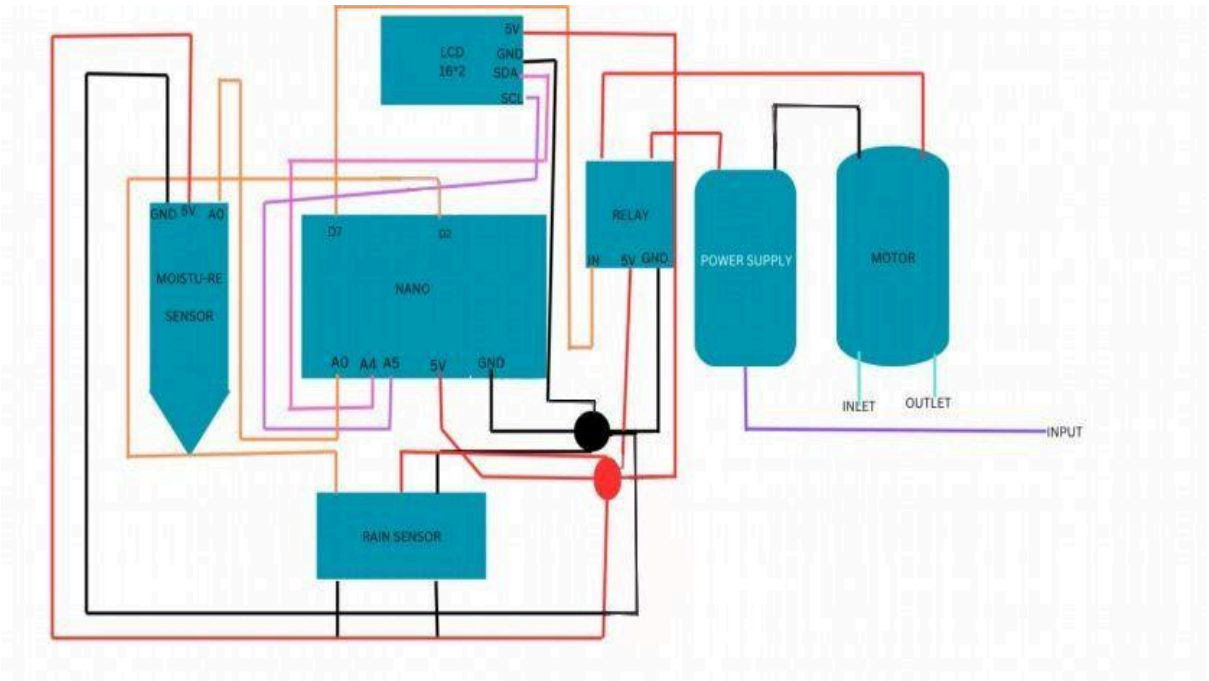
The core of the system involves a capacitive soil moisture sensor interfaced with an ESP8266 microcontroller, which serves as the central processing unit. The microcontroller collects data from the sensor and processes it to determine the soil's moisture content. When watering is required, the system activates a water pump through a relay module, delivering water directly to the soil. This autonomous functionality reduces manual intervention while ensuring optimal plant health.

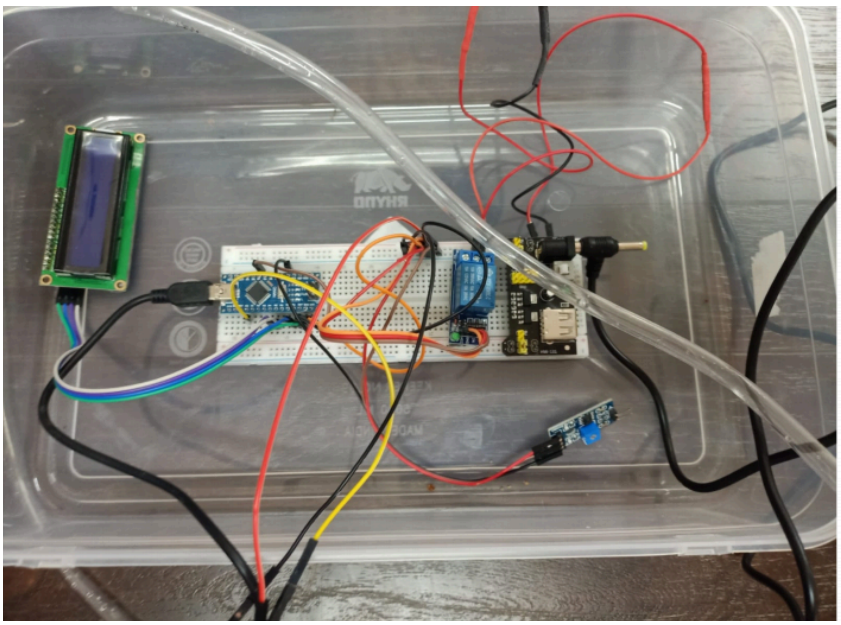
The primary goal of this project is to develop a reliable and user-friendly automated watering system that optimizes water consumption and promotes sustainable irrigation practices. By reducing water wastage and ensuring consistent moisture levels, this project addresses critical issues related to water conservation and efficient agricultural management.

Components

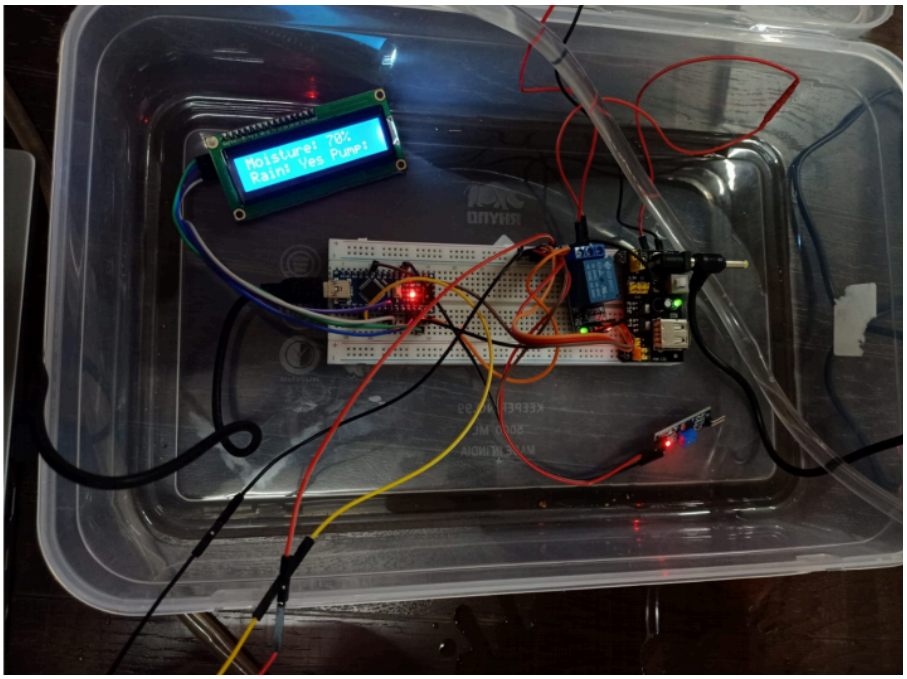
- Arduino nano
- Raindrop sensor module
- DC water pump
- Capacitive Soil moisture sensor
- 5V 1-channel Relay module
- 16 x 2 LCD display
- Python IDE

Block Diagram





automated watering system



automated watering system (on condition)